# Supplementary Material

Physics-Guided Probabilistic Glucose Forecasting in Type 1 Diabetes: An Integrity-Checked Multi-Horizon Evaluation with a Stable Patient-Specific Insulin-Sensitivity Estimate

## Notation, Variables, and Terminology

Rate constants are expressed per minute unless stated otherwise. A subscript $b$ denotes a basal or fasting quantity, a hat denotes a prediction, bold lowercase symbols denote vectors, and bold uppercase symbols denote matrices. Bracketed reference numbers refer to the reference list of the main text. The patient-parameter vector $\theta_p$ is listed in Table XIV of the main text.

Table S1: Physiological state variables and external inputs

| Symbol | Unit | Name | Meaning in this study |
|---|---|---|---|
| $G(t)$ | mg/dL | Plasma glucose | Glucose concentration forecast by the learned spline and constrained by the mechanistic equation. |
| $G_b$ | mg/dL | Basal glucose | Subject-specific fasting anchor used by the Bergman equilibrium. |
| $G_0$ | mg/dL | Anchor glucose | Last real CGM observation at forecast time $t = 0$. |
| $X(t)$ | $\text{min}^{-1}$ | Remote insulin action | Interstitial insulin effect entering glucose disposal additively with $p_1$. |
| $I(t)$ | $\mu$U/mL | Plasma insulin | Circulating insulin concentration generated by subcutaneous absorption and clearance. |
| $I_b$ | $\mu$U/mL | Basal plasma insulin | Concentration implied by the subject's basal delivery; subtracting it preserves basal equilibrium. |
| $S_1, S_2$ | U | SC insulin states | First and second subcutaneous absorption compartments. |
| $Q_{\text{sto}}$ | g | Stomach carbohydrate | Ingested carbohydrate awaiting gastric emptying. |
| $Q_{\text{gut}}$ | g | Gut carbohydrate | Carbohydrate available for intestinal absorption. |
| $R_a(t)$ | mg/min | Glucose appearance rate | Meal-derived glucose entering the accessible glucose compartment. |
| $u_{\text{ins}}$ | U/min | Insulin input | Basal plus bolus insulin delivery on the five-minute grid. |
| $u_{\text{carb}}$ | g/min | Meal input | Recorded carbohydrate ingestion on the grid. |
| $\mathbf{z}$ | – | Compartment state | $[S_1, S_2, I, Q_{\text{sto}}, Q_{\text{gut}}, X]^\mathsf{T}$. |
| $\mathbf{u}$ | – | Input vector | $[u_{\text{ins}}, u_{\text{carb}}, I_b]^\mathsf{T}$. |

Table S2: Architecture, discretization, and uncertainty notation

| Symbol | Value or dimension | Definition |
|---|---|---|
| $t$ | 0–120 min | Time measured from the forecast anchor. |
| $h$ | 30, 60, 90, 120 min | Forecast horizon. |
| $\Delta$ | 5 min | Common temporal grid step. |
| $B$ | batch dependent | Number of windows processed together. |
| $H$ | 4 | Number of reported forecast horizons. |
| $K$ | 12 | Number of cubic B-spline basis functions. |
| $B_k(t)$ | spline basis | The $k$th clamped, uniformly knotted B-spline basis function. |
| $c_k$ | learned scalar | Spline coefficient emitted by the trajectory head. |
| $G_\theta(t)$ | mg/dL | Anchored learned trajectory $G_0 + \sum_k c_k[B_k(t) - B_k(0)]$. |
| $A, B$ | $6 \times 6$, $6 \times 3$ | Continuous compartment-state and input matrices. |
| $A_d, B_d$ | discrete matrices | Exact zero-order-hold matrices obtained from one augmented matrix exponential. |
| $\theta_p$ | 10 elements | Patient-specific physiological parameter vector in Table XIV of the main text. |
| $g = \sigma(\gamma)$ | initialized near 0.10 | Learned trust gate controlling the mechanistic prior. |
| $\hat{G}_{0.5}$ | mg/dL | Median forecast and reported point prediction. |
| $\hat{G}_q$ | mg/dL | Forecast at quantile level $q \in \{0.1, 0.5, 0.9\}$. |
| $\delta_q(h)$ | mg/dL | Positive softplus-derived distance from the median, preventing quantile crossing. |
| $r(t)$ | dimensionless | Scale-normalized physics residual evaluated at 121 collocation points. |
| $n_{\text{eff}}$ | $\approx N/48$ | Approximate independent sample count after accounting for window overlap. |

Table S3: Training-loss terminology

| Term | Meaning |
|---|---|
| $\mathcal{L}_{\text{Huber}}$ | Horizon-weighted robust point loss for the median forecast; quadratic near zero error and linear for large errors. |
| $\mathcal{L}_{\text{pinball}}$ | Proper quantile loss $\max\{q(y-\hat{y}_q),(q-1)(y-\hat{y}_q)\}$. |
| $\mathcal{L}_{\text{phys}}$ | Mean squared, nondimensionalized Bergman residual over the collocation grid. |
| $\mathcal{L}_{\text{prior}}$ | Weak population-prior penalty that discourages physiologically implausible parameter drift. |
| $\mathcal{L}_{\text{temporal}}$ | Within-subject penalty on disagreement between parameter estimates from adjacent windows. |
| $\lambda_{\text{phys}}$ | Physics weight, adaptive in the relevant ablation arms and optionally multiplied by a curriculum ramp. |
| $s_i = \log \sigma_i^2$ | Stable log-variance parameterization used for learned multi-objective weighting [51]. |

Table S4: Evaluation metrics and clinical terms

| Term | Unit | Definition and interpretation |
|---|---|---|
| MAE | mg/dL | Mean absolute error, $n^{-1}\sum_i \lvert\hat{y}_i - y_i\rvert$. |
| RMSE | mg/dL | Root-mean-square error, emphasizing larger deviations. |
| $R^2$ | – | $1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$. |
| MARD | % | Mean absolute relative difference, with reference glucose in the denominator. |
| Skill | % | $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{persistence}}$. Positive values beat no-change prediction. |
| TIR/TAR/TBR | % | Time in, above, or below the consensus glucose range. The main in-range interval is 70–180 mg/dL [55]. |
| CV ratio | – | $\text{CV}_{\text{pred}}/\text{CV}_{\text{actual}}$; values below one quantify excursion compression. |
| LBGI/HBGI | – | Low/high blood-glucose risk indices after the Kovatchev symmetric risk transform. |
| Coverage | – | Fraction of observations below a stated predictive quantile; calibration requires observed and nominal coverage to agree. |
| Hypo sensitivity | – | Fraction of actual $G < 70$ mg/dL events detected by the forecast or alarm. |
| Precision | – | Fraction of hypoglycemia alarms corresponding to actual events. |
| ICC(1) | – | One-way, single-measure intraclass correlation used for $S_I$ test–retest stability [60]. |
| $\rho$ | – | Spearman rank correlation used for monotonic association with insulin dose [61]. |
| $\text{IG}_i$ | mg/dL | Integrated-gradient attribution of feature $i$ relative to the standardized-mean baseline. |
| Clarke/Parkes EGA | zone | Point-wise clinical error grids originally designed for glucose-measurement error. |
| PRED-EGA | category | Prediction-specific error grid combining level and rate information [69]; reported in the Results of the main text. |

Table S5: Abbreviations and protocol terminology

| Short form | Expansion | Study-specific meaning |
|---|---|---|
| T1D/IDDM | Type 1 diabetes / insulin-dependent diabetes mellitus | IDDM is retained only when describing older physiological literature. |
| CGM | Continuous glucose monitoring | Nominal five-minute glucose sampling in OhioT1DM. |
| SC | Subcutaneous | Route represented by insulin compartments $S_1$ and $S_2$. |
| IOB/COB | Insulin/carbohydrate on board | Derived as $S_1 + S_2$ and $Q_{\text{sto}} + Q_{\text{gut}}$. |
| TDD/ICR | Total daily insulin dose / insulin-to-carbohydrate ratio | External consistency variables for the physiology analysis; ICR is available only for six subjects. |
| PINN/ODE | Physics-informed neural network / ordinary differential equation | Neural training constrained by the physiological differential equations. |
| LOSO | Leave-one-subject-out | Twelve subject-disjoint folds, each withholding all data from one test subject. |
| Official protocol | OhioT1DM temporal holdout | Later data from subjects whose earlier data are available for training; therefore personalized. |
| BGLP | Blood Glucose Level Prediction challenge | OhioT1DM challenge setting; 2018 and 2020 have different test warmup rules. |
| EGA | Error-grid analysis | Clarke and Parkes clinical zone classification. |
| IG/SHAP | Integrated gradients / SHapley Additive exPlanations | IG is used; the legacy single-timestep SHAP procedure was rejected. |
| SD/CI/ICC | Standard deviation / confidence interval / intraclass correlation | Uncertainty and reliability summaries computed at the subject level. |
| A0–A7 | Ablation arms | Controlled variants of physics form, weighting, parameter personalization, and curriculum. |
| ZOH | Zero-order hold | Exact interval propagation under piecewise-constant inputs; also used in the literature for persistence. |